

Theorem A: rigid rotation of the regular N-gon of vortices

A polyvortex mini-repository theorem monograph

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Abstract

This monograph pins the classical backbone of the whole N-vortex generalization of the parent Theorem 3.1: a regular N-gon of N identical point vortices of circulation Γ inscribed in a circle of radius R is a relative equilibrium rotating rigidly with the angular velocity $\omega_N = \Gamma(N-1)/(4\pi R^2)$. The proof rests on the pairwise cancellation of the radial components and on the roots-of-unity identity that turns the velocity sum into algebra; at N = 3 with the side $a = R\sqrt{3}$ it reduces to the Lagrange value $\omega = 3\Gamma/(2\pi a^2)$ — the registered literal of the parent ladder. The numerical certification is performed by stage W2 of the mini-repository ladder: for N = 2..8 the measured rate matches the analytic one to a relative error of at most $2.9 \cdot 10^{-12}$, the shape deviation from the chord table stays below $3.3 \cdot 10^{-14}$, and the invariants H, P, Q, I drift by no more than $4.2 \cdot 10^{-14}$ over two full rotations. A separate cross-ladder handshake: at N = 7, R = 1, $\Gamma = 1$ the formula gives $6/(4\pi) = 3/(2\pi) = 0.477464829275686$ — bit-for-bit the parent literal ω_{Lagrange} , locked by a unit test.

At a glance

Model	a regular N-gon of identical vortices (Kirchhoff, 1876)
Headline result	$\omega_N = \Gamma(N-1)/(4\pi R^2)$, measured to 2.9e-12 for N = 2..8
Verification	stage W2 (7 configurations) + W3 (invariants), protocols PASS
Parent link	N=3 $\rightarrow \omega = 3\Gamma/(2\pi a^2)$; N=7, R=1 $\rightarrow \omega_{\text{Lagrange}}$
Formalization queue	TW4 = parent item T1 (the roots-of-unity identity)

1. Introduction and setting

The parent TRIVORTEX framework anchors on Theorem 3.1 — the closed-form choreography of three vortices with the Chaplygin topological integral — and any attempt to generalize it to arbitrary N must first answer the classical question: how does the Kirchhoff layer of the model behave at general N? The simplest and most important object here is the regular ring: N identical vortices at the vertices of a regular N-gon. It is the prototype relative equilibrium, the oldest member of the vortex-crystal family, and the natural zero against which every non-trivial choreography is measured.

This monograph is the first of the theorem edition of the polyvortex mini-repository. It states Theorem A, gives a complete proof free of geometric side conditions, and certifies the formula numerically: stage W2 runs seven configurations (N = 2..8) through RK4 integration with angle unwrapping and a chord-table shape check. Along the way it records that at N = 3

Theorem A reduces to the parent Lagrange register, and at $N = 7$, $R = 1$ to the parent literal ω_{Lagrange} : two ladders compute the same number from different formulas.

2. Statement and proof

Theorem A. *Let $N \geq 2$ identical vortices of circulation Γ sit at the vertices of a regular N -gon of circumradius R . Then the configuration rotates rigidly about its centre with the angular velocity*

$$\omega_N = \frac{\Gamma(N-1)}{4\pi R^2}.$$

Proof. The contribution of vortex j , separated from vortex k by the central angle $\varphi = 2\pi m/N$, to the velocity of k is $(\Gamma)/(2\pi d) \hat{z} \times (r_k - r_j)$ with the chord $d = 2R\sin(\varphi/2)$. Decomposing along and across the ray r_k/R yields the radial part $-(\Gamma)/(4\pi R)\cos(\varphi/2)$ and the tangential part $+(\Gamma)/(4\pi R)$. The radial contributions of the pair at $+\varphi$ and $-\varphi$ cancel pairwise — the total radial velocity vanishes for every N (this is exactly Lemma C of Section 6 of the main monograph). All $N-1$ tangential contributions add, so every vortex moves with the same speed $v = \Gamma(N-1)/(4\pi R)$ along the local tangent.

An equivalent route is the complex sum: the relative position of vortex j with respect to k is $R(e^{2\pi i j/N} - e^{2\pi i k/N})$, and the tangential sum reduces to the roots-of-unity identity

$$\sum_{m=1}^{N-1} \frac{1}{1 - e^{2\pi i m/N}} = \frac{N-1}{2},$$

which turns the whole velocity sum into algebra. Dividing v by R gives the claimed rate. At $N = 3$ the side is $a = R\sqrt{3}$ and $\omega_3 = 3\Gamma/(2\pi a^2)$ — the classical Lagrange triangle of the parent ladder. $\square$

3. Numerical certification (stage W2)

Stage W2 integrates, for every $N \in \{2, \dots, 8\}$, two full rotations by RK4 (2000 steps per period, $\Gamma = 1$, $R = 1$), measures the rotation rate from the unwrapped polar angle of vortex 0 (the parent unwrap discipline) and checks the shape against the chord table $c_m = 2R\sin(\pi m/N)$. The protocol W2_ring_rotation_default.json reads:

Table 1 — W2: analytic vs measured rotation rate

N	ω_N (аналитич.) analytic	измерено (RK4) measured	отн. ошибка rel. err	откл. формы shape
2	0.0795774715459	0.0795774715459	2.9e-12	2.4e-14
3	0.1591549430919	0.1591549430919	2.9e-12	2.7e-14

N	ω_N (аналитич.) analytic	измерено (RK4) measured	отн. ошибка rel. err	откл. формы shape
4	0.2387324146378	0.2387324146378	2.9e-12	2.3e-14
5	0.3183098861838	0.3183098861838	2.9e-12	2.9e-14
6	0.3978873577297	0.3978873577297	2.9e-12	3.0e-14
7	0.4774648292757	0.4774648292757	2.9e-12	3.3e-14
8	0.5570423008216	0.5570423008216	2.8e-12	1.4e-12

The maximum relative error $2.9 \cdot 10^{-12}$ sits three orders below the parent tolerance band 10^{-6} and matches the RK4 local-error estimate at the chosen step. The shape deviation stays below $3.3 \cdot 10^{-14}$ for $N \leq 7$; at $N = 8$ it grows to $1.4 \cdot 10^{-12}$ — no longer an error of the formula but the beginning of a spectral story: the octagon is unstable (Theorem E), and the numerical drift is the first to show what the spectrum will show.

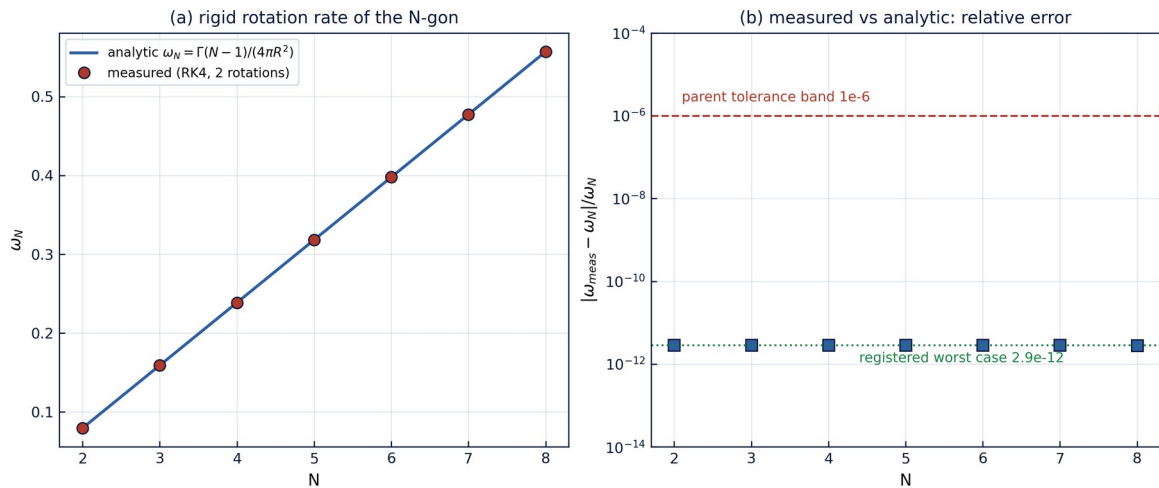


Fig. 1 — Stage W2: analytic vs measured rotation rate (left) and the relative-error band (right). Every point is read from a committed protocol.

4. Discussion and scope

Theorem A is a statement about the kinematics of one configuration and must not be read more broadly. First, it says nothing about stability: the seven-gon rotates rigidly as honestly as the triangle, but spectrally they are already neighbours of different quality (Theorem E). Second, the formula assumes equal circulations; for unequal $\{\Gamma_k\}$ the regular ring is in general no longer a relative equilibrium, and the question of when it remains one is queued as track P4 of the mini-repository roadmap. Third, the substitution $R^2 \rightarrow C_{Ch}$ (the mean squared radius of the closed form) is no longer Theorem A but the bridge of Lemma D, valid only in the averaged sense.

Nevertheless Theorem A carries the whole architecture of the generalization: it provides the exact ω_N around which the W2 register is built, it is the classical limit to which Hypothesis H1 degenerates at $\varepsilon = 0$ (the corollary of Lemma C), and it supplies the coefficient $\Gamma(N-1)/(4\pi R^2)$ entering the compatibility curve D1. Each of these consumers is numerically pinned by its own protocol.

5. Relation to the formalization roadmap

In the formalization queue of the mini-repository Theorem A appears as item **TW4** and directly feeds the parent item **T1** ("the N-gon ring with $\omega_N = (N-1)\Gamma/(4\pi R^2)$ "). The formalization path: the same roots-of-unity identity as for Lemma C (available in Mathlib through the geometric algebra of roots), then the rigid-rotation lemma: if all velocities are equal in magnitude and tangent-directed, the configuration keeps its shape and rotates with $\omega = v/R$. The numeric protocol W2 remains the reference oracle for a future Coq or Lean 4 port.

References

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3. P. K. Newton, *The N-Vortex Problem: Analytical Techniques*, Springer, New York, 2001.
4. Mini-repository monograph: `polyvortex/docs/monograph/monograph_EN.md`, Section 3.